

MARAPPA REDDY CHURNIKA

Data Engineer | Software Engineer | Full-Stack Developer | AI/ML Engineer

Buffalo, NY | +1 716-440-3872 | churnika13@gmail.com | linkedin.com/in/churnika | github.com/churnika67

Professional Summary

Graduate Data Science student with hands-on experience in data engineering, analytics, software engineering, full-stack development, and AI/ML. Built scalable data pipelines, REST APIs, ML workflows, and web/mobile applications using Python, SQL, Spark, Airflow, React, FastAPI, PostgreSQL, PyTorch, and Docker.

Technical Skills

Languages: Python, SQL, JavaScript, TypeScript, Java, C++, HTML, CSS

Data Engineering/Analytics: Airflow, Spark, Iceberg, ETL/ELT, PostgreSQL, Pandas, NumPy, Databricks, Data Validation, Feature Engineering

AI/ML: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, MLflow, Optuna, LLMs, Generative AI, Computer Vision

Full Stack/DevOps: React, React Native, Next.js, FastAPI, REST APIs, Supabase, SQLite, Docker, Git, GitHub Actions, Linux, AWS S3

Experience

University at Buffalo, SUNY

Research Assistant – Generative AI & Data Workflows

Buffalo, NY

Aug 2025 – Present

- Develop Python workflows for data preprocessing, validation, automation, and evaluation of structured and unstructured AI outputs.
- Evaluate LLM responses, identify failure cases, refine prompts, and document reproducible experiments with research teams.

Tirumala Tirupati Devasthanams

Software Engineering Intern

Tirupati, India

Jun 2024 – Aug 2024

- Tested enterprise web applications and debugged frontend, backend, API, and data-flow issues while documenting defects and validating releases.

Slash Mark

Python Intern

Remote

Oct 2023 – Feb 2024

- Developed reusable Python programs for automation, data processing, analysis, debugging, and algorithmic problem-solving using Git workflows.

Selected Projects

Citi Bike Demand Prediction Pipeline

Python, Airflow, Spark, PostgreSQL, ML

- Engineered a Bronze–Silver–Gold pipeline using Airflow, Spark, Iceberg, PostgreSQL, FastAPI, and Docker for station-level demand prediction; achieved **50.9% lower cumulative operational penalty** than the naive baseline in the documented evaluation window.

Evently – Mobile Event Discovery Platform

React Native, TypeScript, Supabase

- Built 10+ user flows including event discovery, search/filtering, recommendations, communities, saved events, RSVP/ticketing, event creation, and profiles using reusable mobile components.

Telco Customer Churn ML System

Python, Scikit-learn, XGBoost, MLflow

- Built an end-to-end ML workflow and compared Logistic Regression, Random Forest, SVC, and XGBoost across 16 configurations with experiment tracking and FastAPI/Streamlit serving.

Real-Time Weather Application

React, FastAPI, SQLite, REST APIs

- Developed a full-stack application integrating live weather APIs with location search, geolocation, five-day forecasts, CRUD persistence, validation, and JSON/CSV exports.

Cognitive Stress Identification

Python, TensorFlow, OpenCV

- Built a real-time computer vision pipeline combining emotion recognition, eyebrow displacement, and blink-pattern analysis for multi-signal stress assessment.

Education

University at Buffalo, SUNY

M.S. in Engineering Science – Data Science

Buffalo, NY

Expected Dec 2026

Amrita Vishwa Vidyapeetham

B.Tech. in Computer Science and Engineering

Coimbatore, India

May 2025